Visualizing optimal habitat zones for potential aquaculture sites

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Objective

Determine which Exclusive Economic Zones (EEZs) on the West Coast of the US are best suited to developing marine aquaculture for several species of oysters and Mediterranean mussels (*Mytilus galloprovincialis*) by creating a) an initial workflow and b) a generalizable workflow function such that new species can be evaluated for suitable habitat areas. Suitable locations will be determined based on range of suitable sea surface temperature (SST) and depth values for the species.

Load packages

```
library(tidyverse)
library(spDataLarge)
library(sf)
library(tmap)
library(gt)
library(viridisLite)
library(terra)
library(stars)
library(knitr)
```

Prepare data

```
# Load EEZ data
eez <- st_read("/Users/courtneylorey/Desktop/GitHub/eds223/eds223-aquaculture/
data/wc_regions_clean.shp")

# Load bathymetry data
depth <- rast("/Users/courtneylorey/Desktop/GitHub/eds223/eds223-aquaculture/
data/depth.tif")

# Load SST rasters
sst8 <- rast("/Users/courtneylorey/Desktop/GitHub/eds223/eds223-aquaculture/
data/average_annual_sst_2008.tif")
```

```
sst9  <-  rast("/Users/courtneylorey/Desktop/GitHub/eds223/eds223-aquaculture/data/average_annual_sst_2009.tif")
sst10  <-  rast("/Users/courtneylorey/Desktop/GitHub/eds223/eds223-aquaculture/data/average_annual_sst_2010.tif")
sst11  <-  rast("/Users/courtneylorey/Desktop/GitHub/eds223/eds223-aquaculture/data/average_annual_sst_2011.tif")
sst12  <-  rast("/Users/courtneylorey/Desktop/GitHub/eds223/eds223-aquaculture/data/average_annual_sst_2012.tif")
```

```
# Combine SST rasters into a raster stack
sst_merged <- c(sst8, sst9, sst10, sst11, sst12)

# Check that datasets have the same crs

# SST vs. depth crs
if(crs(sst_merged) == crs(depth)) {
  print("Coordinate reference systems match")
} else {
  warning("Updating coordinate reference systems to match")
  depth <- project(depth, sst_merged, method = "bilinear")
  if (crs(sst_merged) == crs(depth)) {
    print("Coordinate reference systems have been updated to match")
  }
}
```

```
[1] "Coordinate reference systems have been updated to match"
```

```
# Depth vs. EEZ crs
if(st_crs(depth) == st_crs(eez)) {
  print("Coordinate reference systems match")
} else {
  warning("Updating coordinate reference systems to match")
  eez <- st_transform(eez, crs(depth))
  if (crs(depth) == crs(eez)) {
    print("Coordinate reference systems have been updated to match")
  }
}
```

```
[1] "Coordinate reference systems match"
```

Process data

```
# Create single raster of average SST, convert from Kelvin to Celsius
sst_mean <- mean(sst_merged, na.rm = TRUE)

sst_c <- sst_mean - 273.15

# Crop depth raster to match the extent of the SST raster
depth_crop <- crop(depth, ext(sst_c))

# Resample depth to match the resolution of sst_c
resample_depth <- resample(depth_crop, y = sst_c, method = "bilinear")

# Check that resample_depth and sst_c match in resolution, extent, and crs

# Resolution
if(all(res(resample_depth) == res(sst_c))) {
  print("Resolution matches")
} else {
  warning("Resolutions do not match")
}
```

```
[1] "Resolution matches"
```

```
# Extent
if(ext(resample_depth) == ext(sst_c)) {
  print("Extent matches")
} else {
  warning("Extents do not match")
}
```

```
[1] "Extent matches"
```

```
# CRS
if(crs(resample_depth) == crs(sst_c)) {
  print("CRS matches")
} else {
  warning("CRS does not match")
}
```

```
[1] "CRS matches"
```

```r
# Additional verification: stack sst_c and resample_depth rasters
sst_depth <- c(sst_c, resample_depth)
```

Initial workflow

Find suitable locations

```r
# Create depth matrix
rcl_depth <- matrix(c(
   -Inf, -70,  0,
   -70,  0,  1,
   0, Inf, 0
), ncol = 3, byrow = TRUE)

# Create SST matrix
rcl_SST <- matrix(c(
   -Inf, 11,  0,
   11,  30,  1,
   30, Inf, 0
), ncol = 3, byrow = TRUE)

# Reclassify depth
suitable_depth <- classify(resample_depth, rcl = rcl_depth)

# Reclassify temp
suitable_temp <- classify(sst_c, rcl = rcl_SST)

# Combine suitable zone rasters
suitable_zones <- c(suitable_depth, suitable_temp)

# Find suitable locations by multiplying suitable zones
suitable_locations <- lapp(suitable_zones, fun = function(sst, depth) sst *
depth)
```

Determine the most suitable EEZs

```r
# Rasterize EEZ data
eez_raster <- rasterize(eez, suitable_locations, field = "rgn_id")

# Calculate the area of each cell
cell_sizes <- cellSize(suitable_locations, unit = "km", mask = TRUE)

# Make sure only cells classified as '1' contribute to area totals
```

```r
suitable_area <- mask(cell_sizes, suitable_locations, maskvalues = 0)

# Find the suitable area for each EEZ region
area_per_eez <- zonal(suitable_area, eez_raster, fun = "sum", na.rm = TRUE)
```

Visualize

Table

```r
area_per_eez |>
  mutate(
    Region = c("Oregon", "Northern California", "Central California", "Southern
California", "Washington")
  ) |>
  knitr::kable(
    col.names = c("Region ID", "Area $(km^2)$", "Region"),
    digits = 0,
    caption = "Suitable Oyster Habitat Area by West Coast EEZ Region"
  )
```

Region ID	Area (km^2)	Region
1	1396	Oregon
2	324	Northern California
3	4374	Central California
4	3330	Southern California
5	3092	Washington

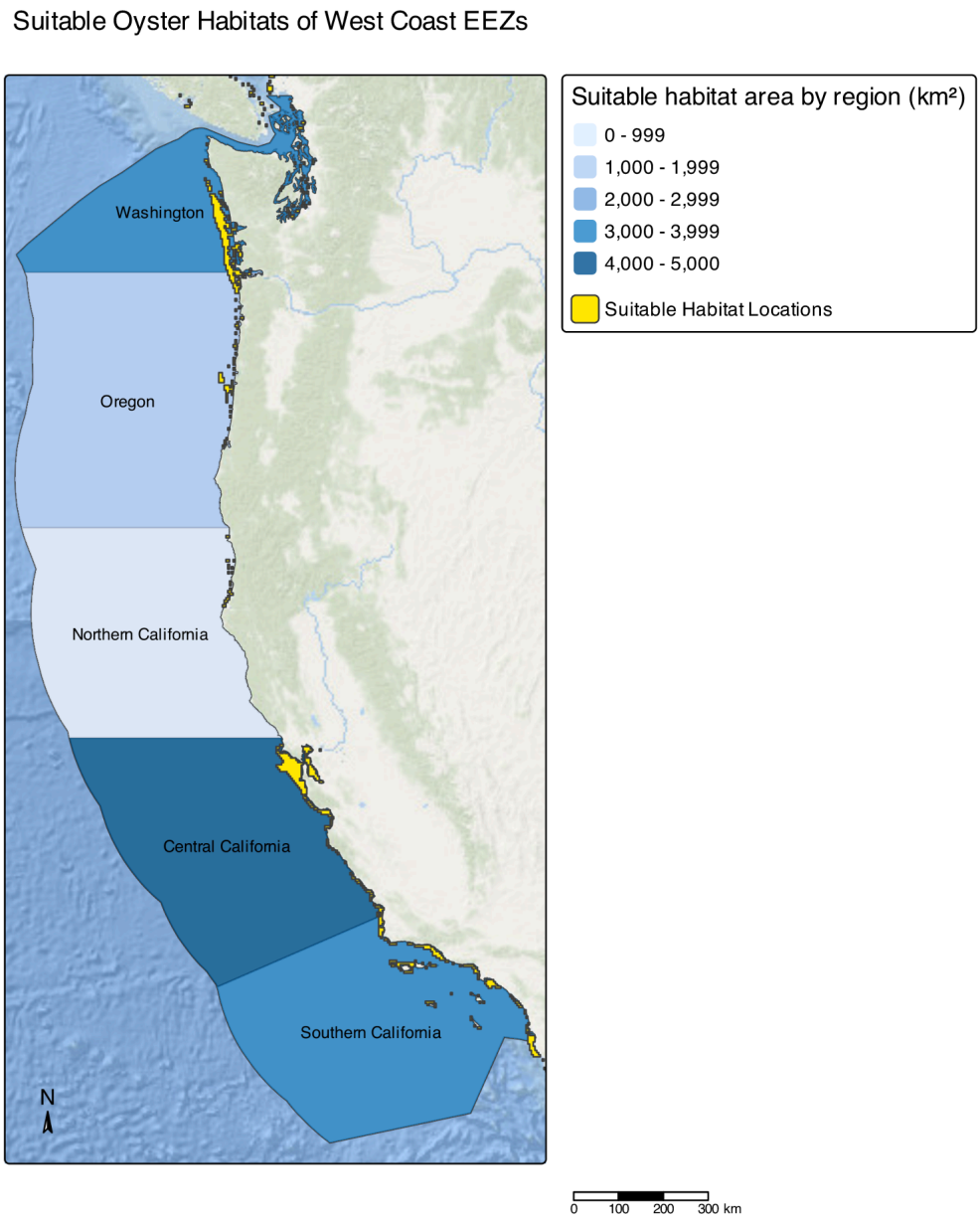
Map

```r
# Add EEZ polygons back in
map <- left_join(eez, area_per_eez, by = "rgn_id")

# Convert suitable area spatraster to polygons for plotting
area_poly <- as.polygons(suitable_area, dissolve = TRUE)

# Visualize
tm_shape(map) +
  tm_basemap("Esri.OceanBasemap") +
  tm_polygons() +
  tm_fill(
    fill = "area",
```

```
    fill_alpha = 0.8,
     fill.legend = tm_legend(title = "Suitable habitat area by region (km²)",
position = tm_pos_out("right"))
  ) +
  tm_text(text = "rgn", size = 0.6) +
  tm_title(text = "Suitable Oyster Habitats of West Coast EEZs", size = 1.5) +
  tm_compass(position = c("left", "bottom"), size = 0.8) +
  tm_scalebar(position = tm_pos_out("right", "bottom")) +
tm_shape(area_poly) +
  tm_polygons(fill = "#FFEA00") +
  tm_add_legend(
    type = "polygons",
    labels = "Suitable Habitat Locations",
    fill = "#FFEA00",
    position = tm_pos_out("right")
  )
```



Oyster interpretation

From this table, we can determine that Central California has the most suitable oyster habitat when compared to the other four regions. By contrast, Northern California has the least suitable oyster habitat.

The yellow areas on the map represent the suitable habitat locations for oysters, where sea surface temperature is between 11-30°C and depth is 0-70 meters below sea level. The map shows

that these areas are concentrated in the EEZs of Central California, Southern California, and Washington.

Generalizable workflow

Build the function

```
suitable_habitat_area <- function(
  min_sst,
  max_sst,
  min_depth,
  max_depth,
  species_name
) {
# Create SST matrix
rcl_SST <- matrix(c(
   -Inf, min_sst,  0,
   min_sst,  max_sst,  1,
   max_sst, Inf, 0
), ncol = 3, byrow = TRUE)

# Reclassify SST
suitable_temp <- classify(sst_c, rcl = rcl_SST)

# Create depth matrix
rcl_depth <- matrix(c(
   -Inf, max_depth,  0,
   max_depth,  min_depth,  1,
   min_depth, Inf, 0
), ncol = 3, byrow = TRUE)

# Reclassify depth
suitable_depth <- classify(resample_depth, rcl = rcl_depth)

# Combine depth and temp into a suitable zone raster
suitable_zones <- c(suitable_depth, suitable_temp)

# Find suitable locations by multiplying suitable zones
suitable_locations <- lapp(suitable_zones, fun = function(sst, depth) sst *
depth)

# Rasterize EEZ data
eez_raster <- rasterize(eez, suitable_locations, field = "rgn_id")
```

```
# Calculate the area of each cell
cell_sizes <- cellSize(suitable_locations, unit = "km", mask = TRUE)

# Make sure only cells classified as 1 contribute to area totals
suitable_area <- mask(cell_sizes, suitable_locations, maskvalues = 0)

# Find the suitable area for each EEZ region
area_per_eez <- zonal(suitable_area, eez_raster, fun = "sum", na.rm = TRUE)

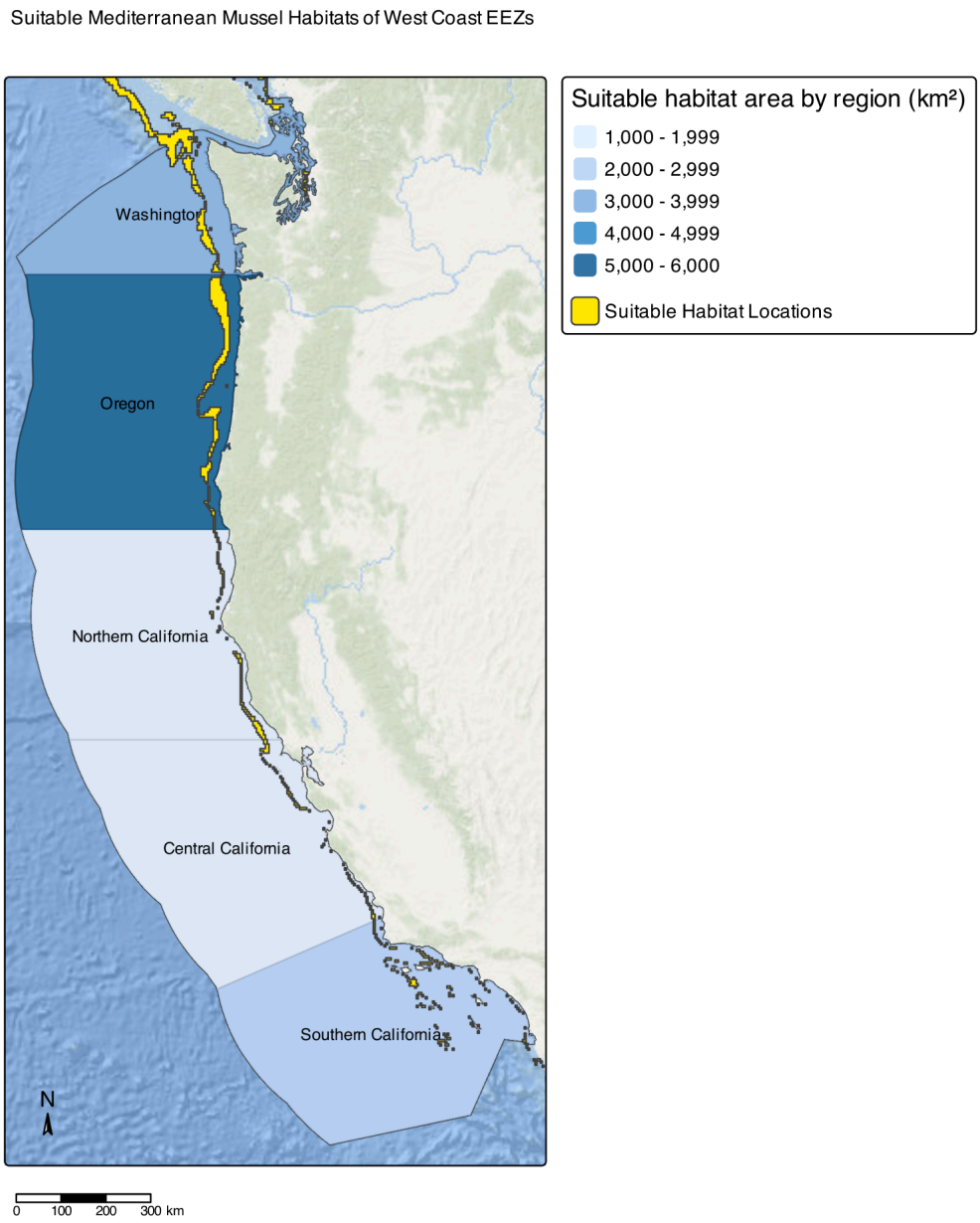
# Add EEZ polygons back in for plotting
map <- left_join(eez, area_per_eez, by = "rgn_id")

# Convert suitable area SpatRaster to polygons for plotting
area_poly <- as.polygons(suitable_area, dissolve = TRUE)

# Visualize
tm_shape(map) +
  tm_basemap("Esri.OceanBasemap") +
  tm_polygons() +
  tm_fill(
    fill = "area",
    fill_alpha = 0.8,
     fill.legend = tm_legend(title = "Suitable habitat area by region (km²)",
position = tm_pos_out("right"))
  ) +
  tm_text(text = "rgn", size = 0.6) +
  tm_title(paste("Suitable", species_name, "Habitats of West Coast EEZs"), size
= 1.5) +
  tm_compass(position = c("left", "bottom"), size = 0.8) +
  tm_scalebar(position = tm_pos_out("center", "bottom")) +
tm_shape(area_poly) +
  tm_polygons(fill = "#FFEA00") +
  tm_add_legend(
    type = "polygons",
    labels = "Suitable Habitat Locations",
    fill = "#FFEA00",
    position = tm_pos_out("right")
  )
}
```

Use the function to visualize suitable habitats for Mediterranean mussels

```
suitable_habitat_area(10, 20, -125, -200, "Mediterranean Mussel")
```



Mediterranean mussel (*Mytilus galloprovincialis*) interpretation

The yellow areas on the map represent the suitable habitat locations for Mediterranean mussels, where sea surface temperature is between 10-20°C and depth is 125-200 meters below sea level. The map shows that the West Coast EEZ with the most suitable habitat is Oregon.

Note: The SeaLifeBase page on *Mytilus galloprovincialis* lists their depth range as 125 - 200 m, which seems very deep compared to other sources (where it is listed as having a depth range of 0-40 m), and considering the fact that it is an inter- and subtidal species. FAO For the purposes

of this exercise I chose to use the information from SeaLifeBase, but it is worth noting that depth information may be misreported on SeaLifeBase.